

JKU
Visual Data
Science Lab

AI and Visualization

Lecture + Lab Introduction and Course Overview

Univ.-Prof. Dr. Marc Streit

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Marc Streit

marc@streit.com

marc-streit.com

TU Graz, Austria | 2002 - 2012
MSc & PhD in Computer Science

JKU Linz, Austria | since 2012
Full Professor for Visual Data Science

Harvard Medical School, US | 2012
Visiting Scholar

Harvard School of Eng., US | 2014
Visiting Professor, Fulbright Scholar

Imperial College London | 2016 - 2023
Lecturer

datavisyn | since 2017
CEO > CSO and Co-Founder

Course Team

Responsible for answering all lab specific questions

Amal Alnouri
Instructor



Dr. Andreas Hinterreiter
Instructor and Lecturer

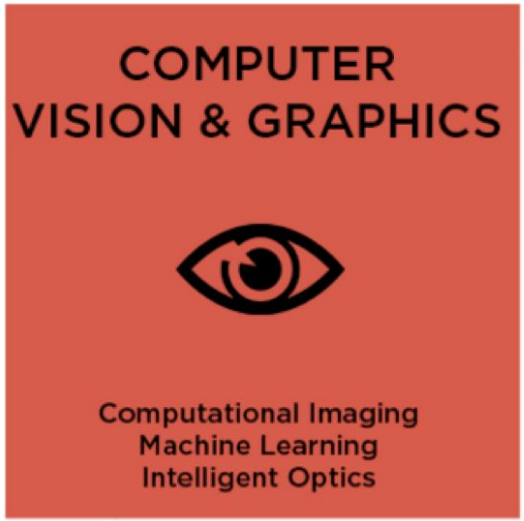


Ahmed Mansour
Constantin Rebasso
Florian Holeczek
Gaser Eldib
Giovanni Filomeno
Teaching Assistants

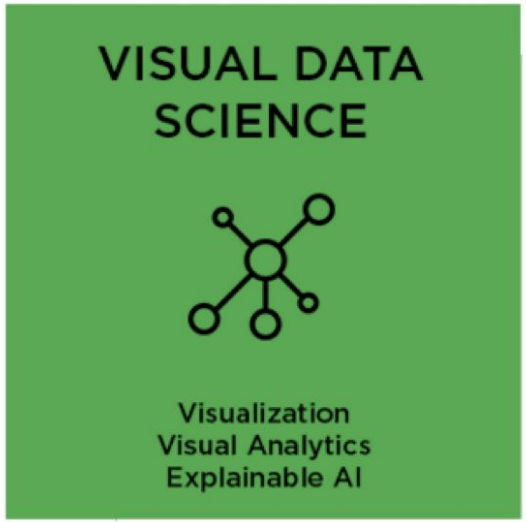
JKU Institute of Computer Graphics

Computer Science Building, 3rd floor

www.jku.at/cg



Prof. Bimber



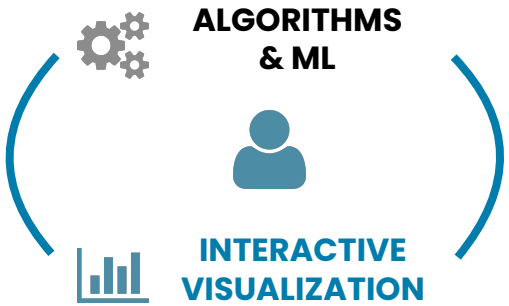
Prof. Streit



Prof. Wallner

JKU Visual Data Science Lab

https://jku-vds-lab.at/



VISUAL ANALYTICS
VISUAL DATA SCIENCE

Visualization

Biological Data Visualization

Explainable AI

What’s your background?

Course Overview

General Information (1/2)

- Lecture
  - VL, 1.5 ECTS
  - Time: Thursday, 10:15-11:45 (starting on time!)
  - Location: Lecture hall HS 5 + Zoom live stream + recording

- Lab
  - UE, 1.5 ECTS
  - 9 KUSSS groups
  - Time: Thursday, 12:00-13:30, 13:45-15:15; Friday, 08:30-10:00
  - Location: S3 055 + Zoom live stream + recording (if needed)

General Information (2/2)

- Moodle page
  - https://moodle.jku.at/course/view.php?id=43753
- General point of contact
  - Moodle announcements + forum
  - For individual requests: ai-and-vis-course@jku.at
- No mandatory prerequisites

Lab Organization

- Mix of tutorials + assignments + presentations
- **Groups:** Thursday, 12:00-13:30, 13:45-15:15; Friday 08:30-10:00

- **Sessions and deadlines**
  - See Moodle page
  - Assignment introduction on Thursday next week (15.10.2026) at 12:00-13:30, 13:45-15:15
  - First deadline: Assignment group formation; 21.10.2026

Lab Attendance

- **Attendance in final live presentation session is compulsory!**
  - Either on-site or remote
  - For Zoom: Turn on video and join with full name!

- Attendance in all other sessions is recommended but not compulsory

Lab Assignments

- Group Assignment
  - Groups of 4 students
  - Can be from different KUSSS lab groups

- Investigation of AI model through two phases
  - **Phase I: How does the model behave?**
  - **Phase II: What did the model learn?**

Grading

- Lecture
  - Written On-Site Moodle Exam in Linz, Vienna, and Bregenz
    - Main exam: Thursday, 11.12.2025, 10:15-11:45
    - Re-exam: Thursday, 19.02.2026, 10:15-11:45
- Lab
  - Group Assignment
  - Everyone without a group will be unregistered.
  - Everyone who submits a group will be graded

- Grading key
  - <50 5, <63 4, <75 3, <88 2, >=88 1

Introduction

AI + VISUALIZATION + HUMAN = EXPLAINABLE AI

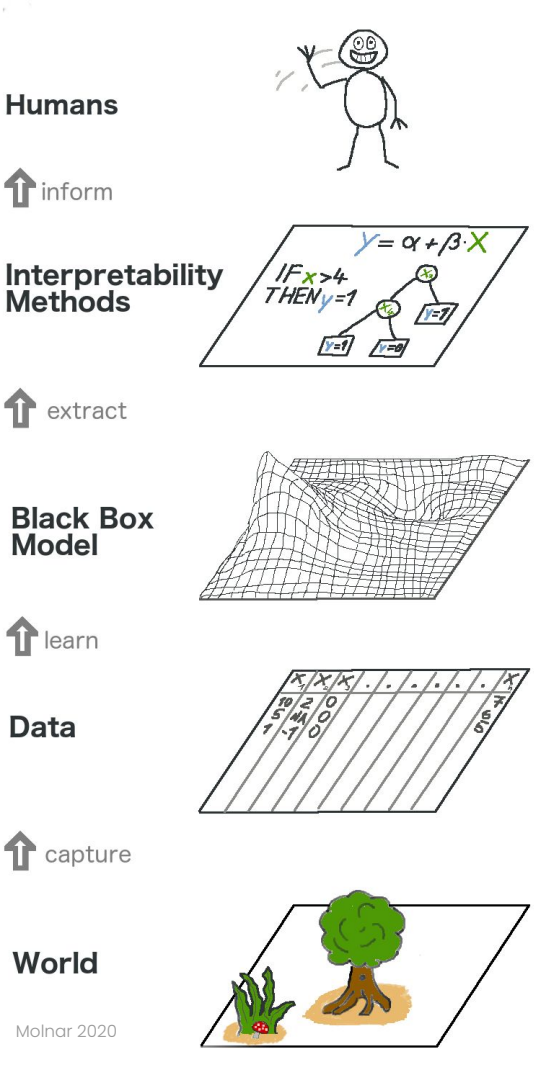


?

The Big Picture of Explainable AI

Various Purposes

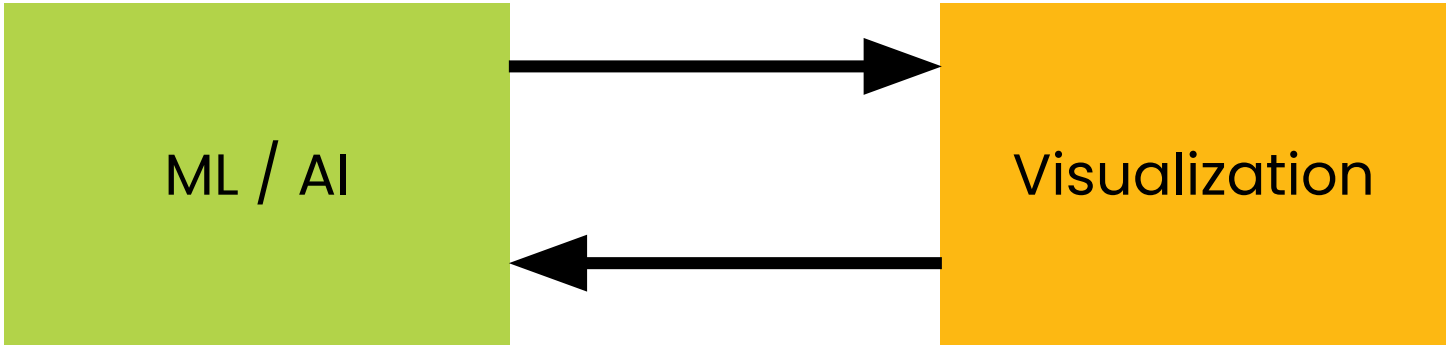
- Interpretability & Explainability
- Debugging & Improving Models
- Comparing & Selecting Models
- Teaching & Understanding Concepts



Vis 4 ML vs. ML 4 Vis

ML 4 Vis

Example: Generating visualizations using AI



Vis 4 ML = Explainable AI (XAI)

Example: Visualizing latent space of a neural network

Disclaimer

I am...

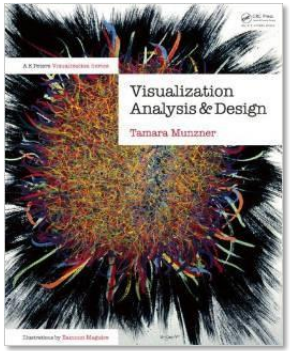
a visualization researcher.

I am NOT...

an ML / AI expert.

Visualization Fundamentals

- **Related courses**
  - Visualization (formerly Visual Analytics)
    VL (364.024) + UE (364.025-364.034), summer term
  - Advanced Interactive Visualization (formerly Information Visualization)
    KV (364.047), summer term
- **Reading**
  - Tamara Munzner,
    Visualization Analysis and Design (VAD)
    CRC Press, 2014. ISBN 1466508914
    http://www.cs.ubc.ca/~tmm/vadbook/

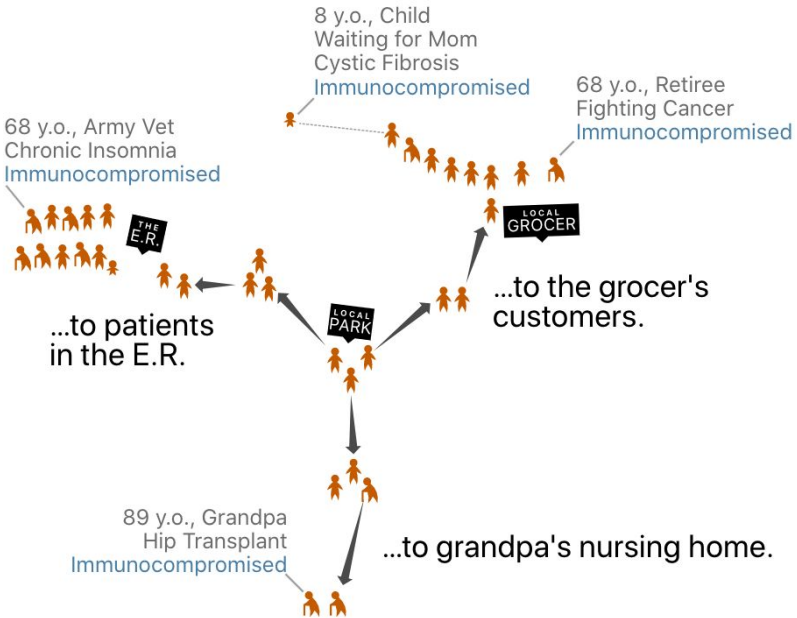


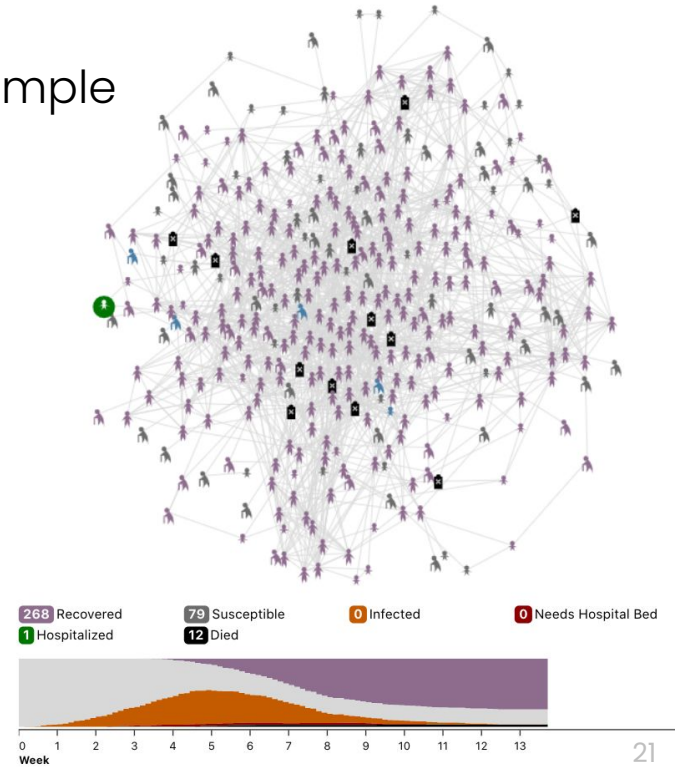


Overview of Topics

Fundamentals & Explaining Algorithms

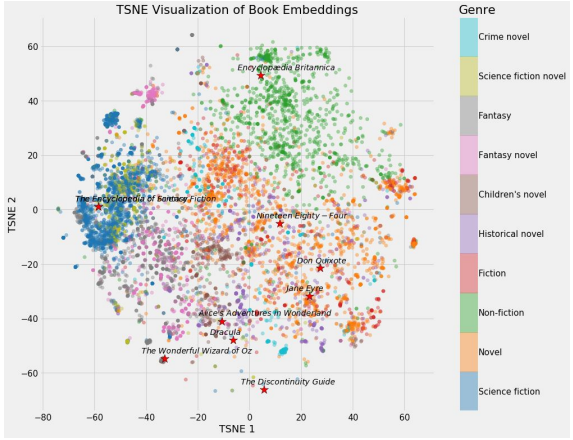
Making Sense of COVID19 Through Simulations
http://www.r2d3.us/covid-19/ - Scrollytelling Example





Explaining Through Projections

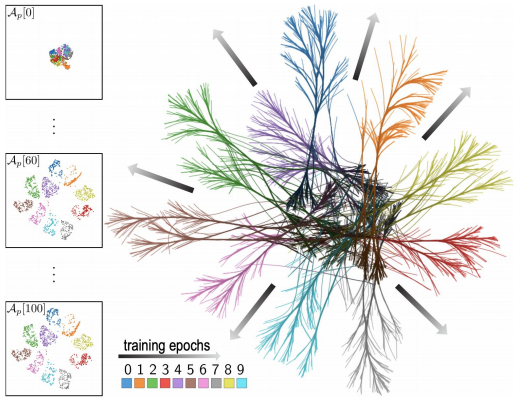
- Dimensionality reduction techniques
  - PCA, UMAP, t-SNE
  - Visualization of embeddings
- Embedding-based trajectories



Embedding of all books on Wikipedia

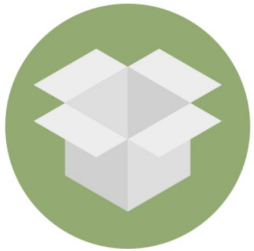


Steinparz et al. 2019 & Hinterreiter et al. 2020



Rauber et al. 2016

Visual Explanation Techniques I & II



Intrinsic interpretability

VS

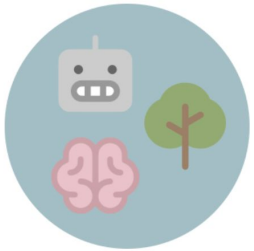
Post-hoc explanation



Global

VS

Local



Model-agnostic

VS

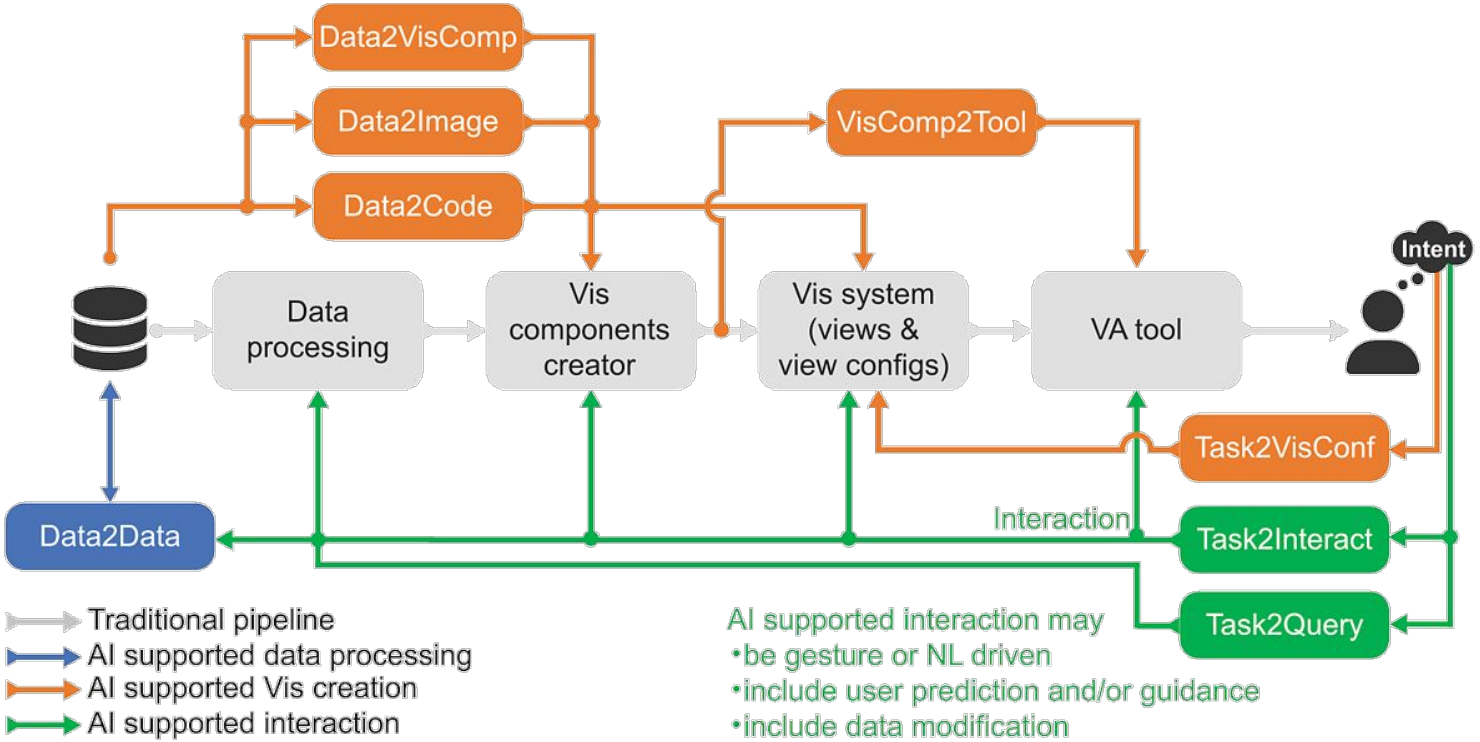
Model-specific







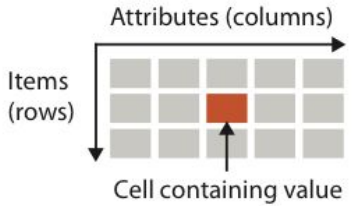
Generative AI & VIS



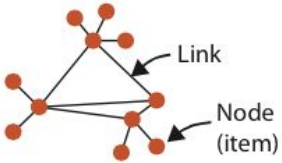
Bühler et al. CG&A, 2025

Excursions to Visualization Principles

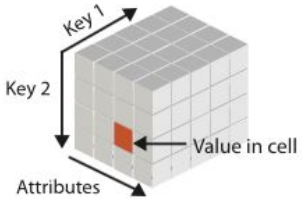
➔ Tables



➔ Networks



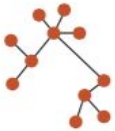
➔ *Multidimensional Table*



➔ *Trees*



➔ **Node–Link Diagrams**
Connection Marks
✓ NETWORKS ✓ TREES



➔ **Adjacency Matrix**
Derived Table
✓ NETWORKS ✓ TREES



➔ **Enclosure**
Containment Marks
✗ NETWORKS ✓ TREES





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Thank you!